

Pearls AQI Predictor

Project Report



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Domain: Data Sciences

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Islamabad, Pakistan

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Islamabad AQI Prediction System

Project Objective

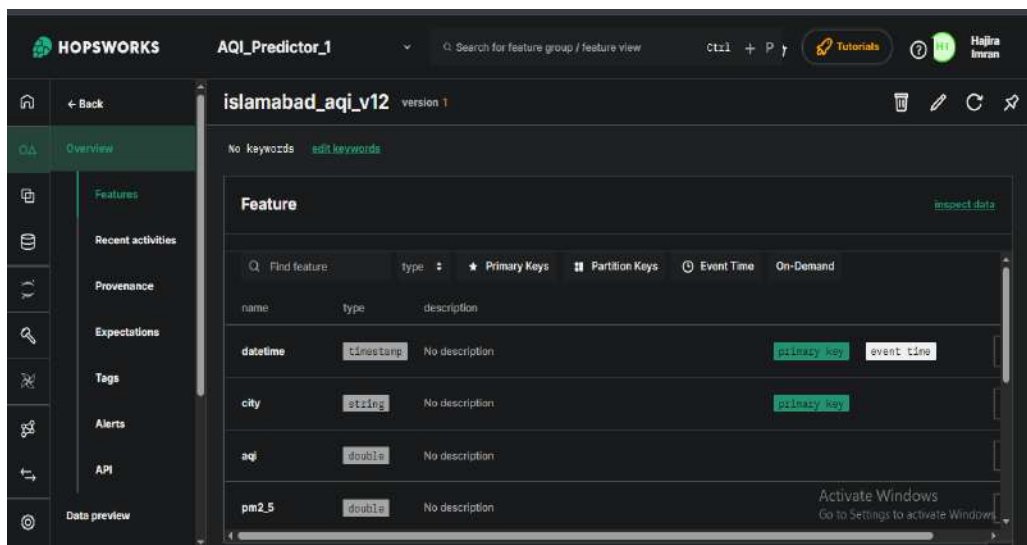
The goal was to build a machine learning pipeline capable of predicting Air Quality Index (AQI) levels for Islamabad using historical and real-time data fetched from the Hopsworks Feature Store.

Problem Definition

Air Quality Index (AQI) prediction is framed as a short-term time-series regression problem with categorical interpretation. The objective is to forecast AQI levels up to three days ahead using historical pollution data, weather conditions, and temporal patterns.

Data Acquisition & Ingestion

- **Source:** Data was retrieved from a remote feature store hosted on Hopsworks.
- **Temporal Scope:** The data spans multiple months, with the most recent records appearing in **January 2026**.



Data Quality Audit & Cleaning

- **Completeness:** A systematic audit confirmed **zero missing values** across all 14 columns, including target variables and engineered features.

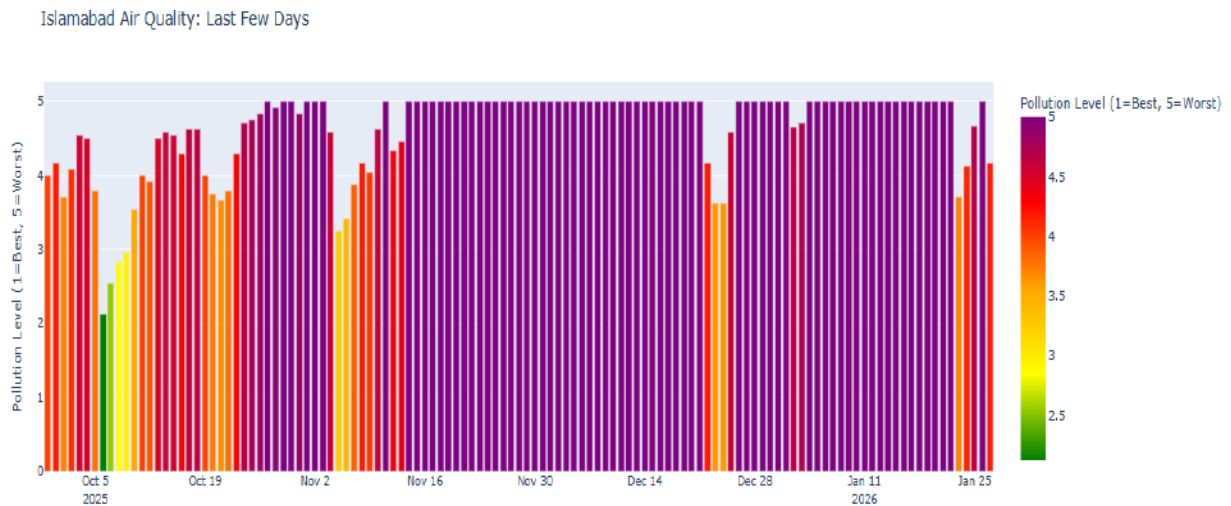
- **Data Consistency:** The non-null count was verified for all features, ensuring stability for model training.
- **Dtypes Verification:** All essential features were confirmed to be in float64 format, ensuring compatibility with mathematical modeling.

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2856 entries, 0 to 2855
Data columns (total 15 columns):
#   Column              Non-Null Count  Dtype
---  -
0   datetime            2856 non-null  datetime64[us, Etc/UTC]
1   city                2856 non-null  object
2   aqi                 2856 non-null  float64
3   pm2_5               2856 non-null  float64
4   no2                 2856 non-null  float64
5   so2                 2856 non-null  float64
6   temperature         2856 non-null  float64
7   humidity            2856 non-null  float64
8   wind_speed          2856 non-null  float64
9   hour                2856 non-null  float64
10  weekday             2856 non-null  float64
11  month               2856 non-null  float64
12  aqi_lag_1           2856 non-null  float64
13  pm2_5_rolling_6h    2856 non-null  float64
14  wind_stagnant        2856 non-null  float64
dtypes: datetime64[us, Etc/UTC](1), float64(13), object(1)
memory usage: 334.8+ KB
```

```
Null values per column:
datetime      0
city          0
aqi           0
pm2_5         0
no2           0
so2           0
temp          0
humidity      0
wind_speed    0
hour          0
weekday       0
month         0
aqi_lag_1     0
pm2_5_rolling_6h 0
wind_stagnant 0
dtype: int64
```

EDA Key Insights

Daily Average AQI Trend



Objective:

To observe how air quality changes on a day-to-day basis and identify short-term pollution trends.

Method:

AQI data was resampled on a daily level, and the average AQI per day was calculated. A color-coded bar chart was used to visualize pollution severity from low (green) to high (purple).

Observations:

Daily AQI levels show noticeable fluctuations, with some days experiencing higher pollution spikes than others.

Modeling Insight:

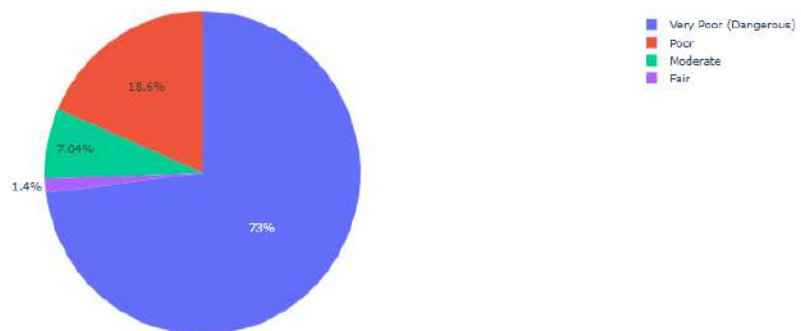
These variations indicate that recent AQI history is important, supporting the use of lag features and time-based variables in the forecasting model.

Conclusion:

AQI is not stable across days, making short-term prediction both necessary and meaningful for early pollution warnings

AQI Category Distribution

Overall Air Quality Distribution in Islamabad

**Objective:**

To understand the overall distribution of air quality conditions by categorizing AQI values into standard pollution levels.

Method:

AQI values were rounded and mapped into five categories: *Good*, *Fair*, *Moderate*, *Poor*, and *Very Poor*. A pie chart was created to visualize the proportion of time the air falls into each category.

Observations:

The distribution shows that air quality is not consistently in the “Good” range, with a noticeable portion of readings falling into *Moderate* and *Poor* categories.

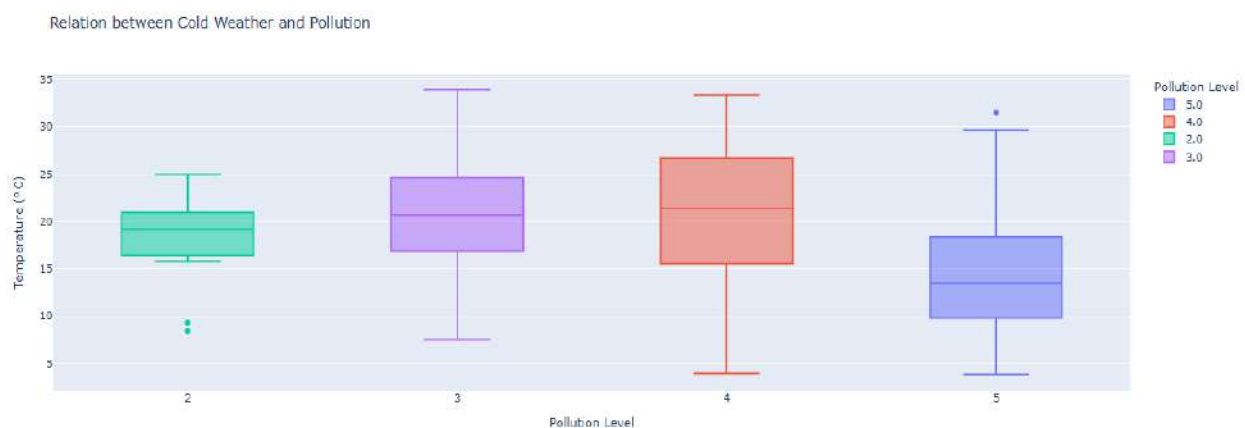
Modeling Insight:

Since AQI frequently reaches unhealthy levels, accurate forecasting is important for early warnings. The imbalance in category distribution also suggests the need to ensure the model performs well during high-pollution events, not just average days.

Conclusion:

Islamabad experiences varying air quality conditions, with a significant share of readings indicating moderate to poor pollution levels, reinforcing the need for predictive monitoring.

Temperature vs AQI Relationship



Objective:

To analyze how temperature influences air pollution levels and determine whether colder weather is associated with higher AQI.

Method:

A box plot was created to compare the distribution of temperature values across different AQI levels. This visualization helps identify whether certain pollution categories tend to occur during specific temperature ranges.

Observations:

Lower temperature ranges appear more frequently in higher AQI categories, suggesting that pollution tends to increase during colder conditions. Warmer temperatures are generally associated with lower AQI levels.

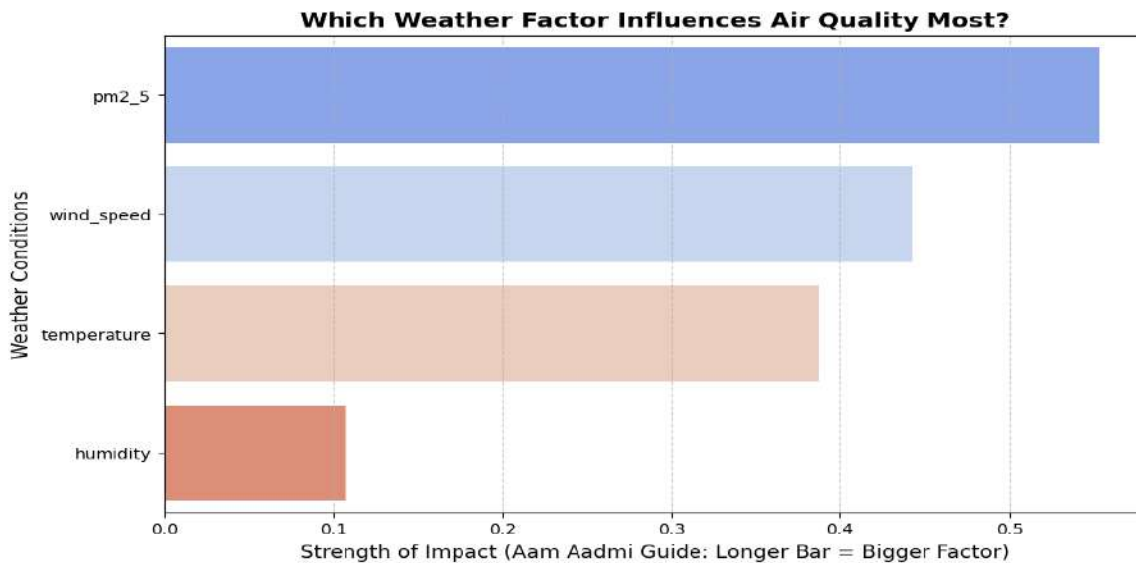
Modeling Insight:

This relationship indicates that **temperature is an important predictive feature** for AQI forecasting. Seasonal and weather-related variables should therefore be included in the model to capture environmental effects on pollution levels.

Conclusion:

The analysis shows a clear link between colder weather and increased pollution, supporting the inclusion of meteorological features such as temperature in the forecasting pipeline.

Weather Factors Impacting AQI



Objective:

To identify which meteorological factors most strongly influence air pollution levels in Islamabad.

Method:

Correlation analysis was performed between AQI and key weather variables: *temperature*, *humidity*, *wind speed*, and *PM2.5*. Absolute correlation values were visualized using a horizontal bar chart to show the strength of each factor's impact.

Observations:

- PM2.5 has the strongest influence on AQI.
- Temperature and wind speed also show moderate effects.
- humidity has a small effect.

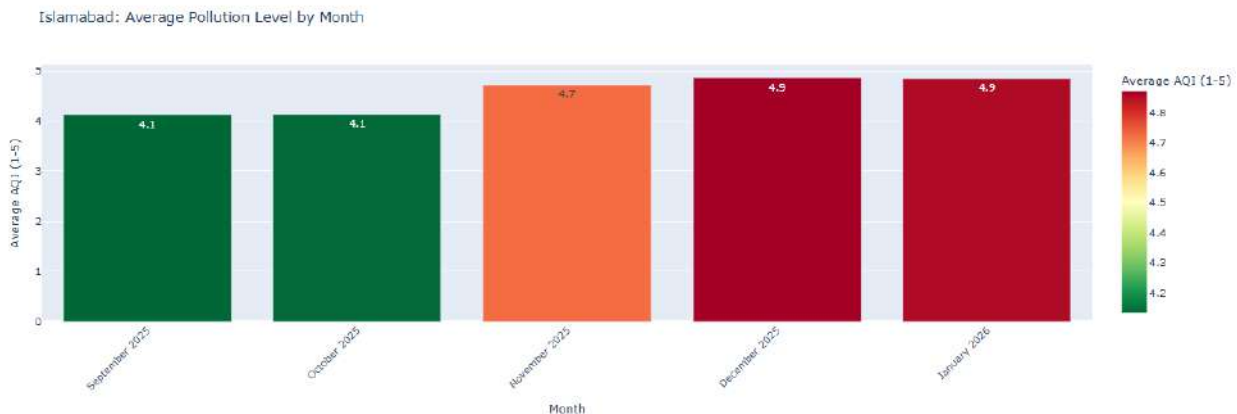
Modeling Insight:

This analysis highlights that **PM2.5 is a critical predictor** for forecasting AQI, and that other weather features like temperature and wind speed should also be included in the model for better accuracy.

Conclusion:

Understanding the relative impact of weather factors helps prioritize features in the prediction pipeline, ensuring that the model captures the most influential environmental variables.

Monthly Average AQI Trends



Objective:

To examine how air quality varies across months and identify seasonal patterns in pollution levels.

Method:

AQI data was grouped by month, and the monthly average AQI was calculated. A color-coded bar chart was created to visualize the variation, with green indicating low pollution and red indicating high pollution.

Observations:

- Certain months show higher average AQI, suggesting seasonal pollution peaks.
- Other months have consistently lower AQI, indicating cleaner air periods.

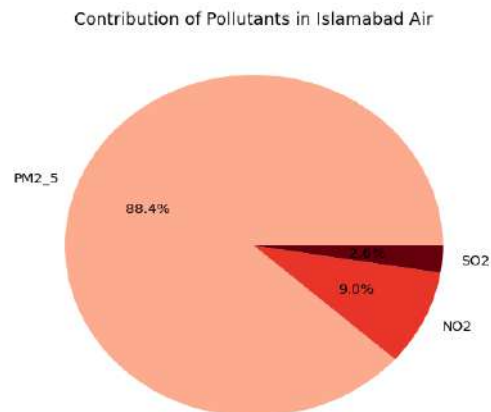
Modeling Insight:

Monthly or seasonal patterns suggest that **time-based features like month or season** should be included in the forecasting model to improve prediction accuracy.

Conclusion:

Air quality in Islamabad exhibits seasonal variation, emphasizing the importance of incorporating temporal features for better short-term AQI forecasts.

Pollutant Contribution to AQI



Objective:

To determine which pollutants (PM_{2.5}, NO₂, SO₂) contribute most to air pollution in Islamabad.

Method:

The average values of key pollutants were calculated and visualized using a pie chart to show their relative contribution to overall air quality.

Observations:

- **PM_{2.5}** is the dominant contributor, followed by NO₂ and SO₂.
- Fine particulate matter (PM_{2.5}) plays a major role in driving AQI levels.

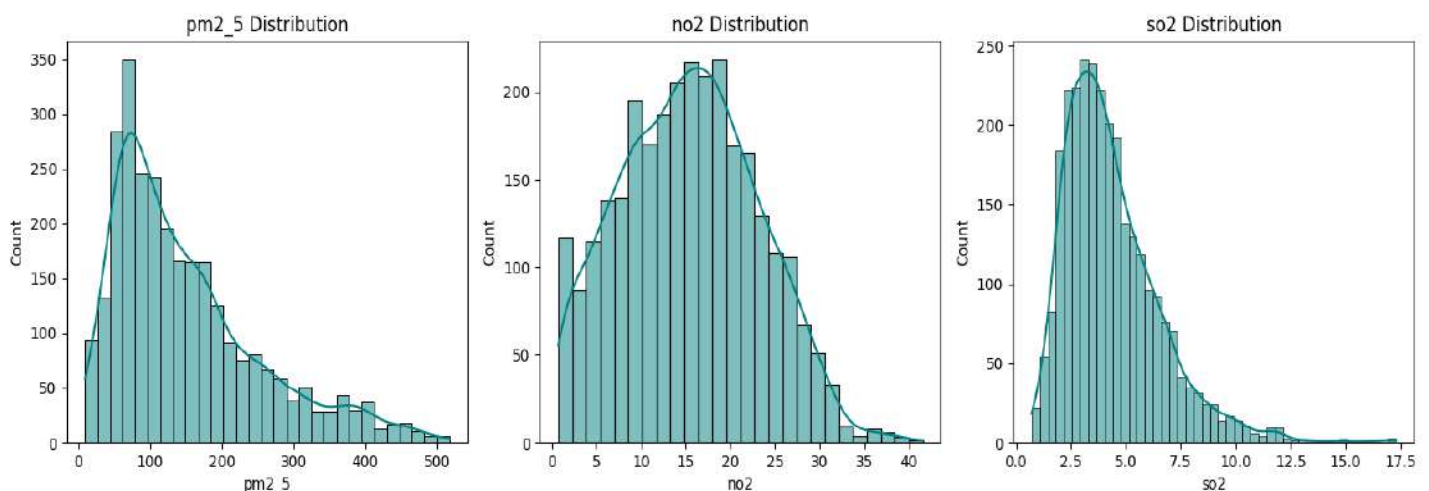
Modeling Insight:

Pollutant concentrations, especially PM_{2.5}, are essential features for AQI prediction. Including these variables ensures the model captures the main drivers of pollution.

Conclusion:

Air quality in Islamabad is heavily influenced by PM_{2.5}, making it a critical factor in forecasting short-term AQI trends

Distribution of Key Pollutants



Objective:

To examine how concentrations of major pollutants (PM_{2.5}, NO₂, SO₂) vary and understand their statistical distribution.

Method:

Histograms with kernel density estimates (KDE) were plotted for each pollutant to visualize the frequency and spread of their values.

Observations:

- PM_{2.5} and NO₂ show a **right-skewed distribution**, indicating occasional high pollution spikes.
- SO₂ concentrations are generally lower and more consistent.
- Pollutant levels vary widely, suggesting that extreme pollution events are possible.

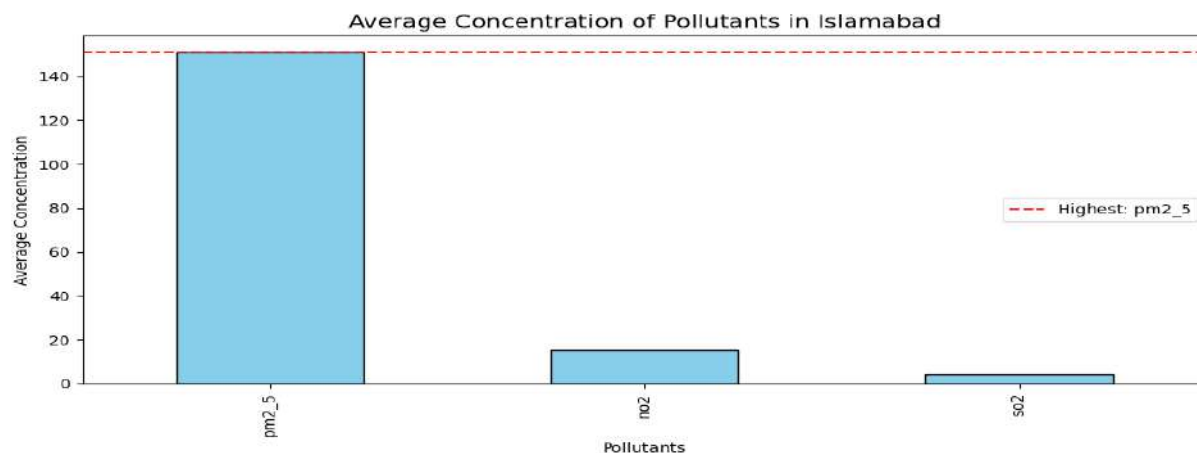
Modeling Insight:

Understanding pollutant distributions helps in **feature scaling, transformation, and outlier handling** for the forecasting model. PM_{2.5} and NO₂, due to their high variability, are particularly important predictors.

Conclusion:

Pollutant distributions reveal the dominant and variable contributors to AQI, guiding effective feature engineering for accurate short-term predictions

Most Critical Pollutant Analysis



Objective:

To identify which pollutant contributes most to air pollution in Islamabad.

Method:

The average concentration of key pollutants (PM2.5, NO₂, SO₂) was calculated and visualized using a bar chart. The pollutant with the highest average was highlighted for emphasis.

Observations:

- **PM2.5** is the most critical pollutant, having the highest average concentration.
- NO₂ and SO₂ contribute less but are still relevant to overall AQI.

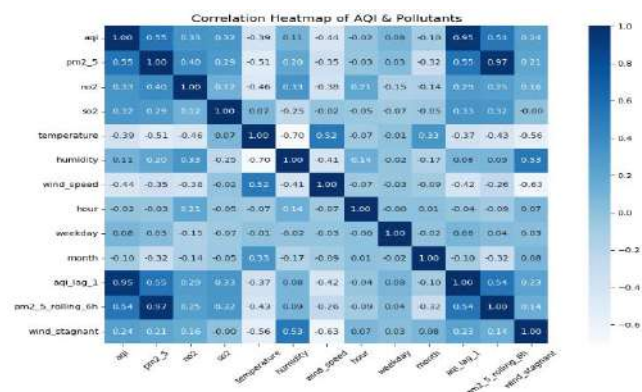
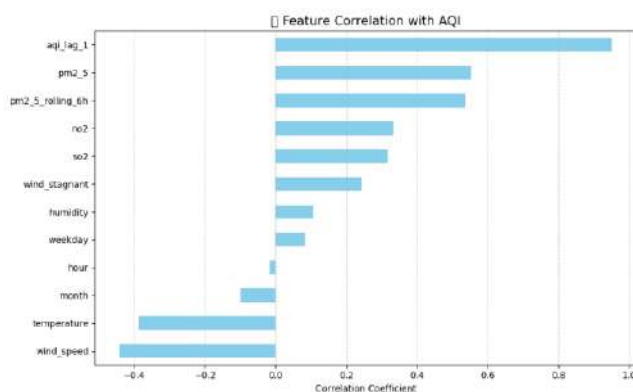
Modeling Insight:

Since PM2.5 dominates pollution levels, it should be a **primary feature** in the AQI forecasting model. Understanding its relative importance helps prioritize feature selection and monitoring efforts.

Conclusion:

PM2.5 is the leading driver of air quality degradation in Islamabad, emphasizing its importance in both prediction models and policy measures.

Correlation Analysis of AQI and Features



Objective:

To identify which features have the strongest relationships with AQI, helping prioritize predictors for the forecasting model.

Method:

- Pearson correlation coefficients were calculated between AQI and all numeric features, including lagged AQI and pollutants.
- Strong correlations ($|\text{corr}| > 0.7$) were highlighted.
- Visualizations included a **correlation heatmap** and a **horizontal bar plot** showing each feature's correlation with AQI.

Observations:

- **aqi_lag_1** (previous hour/day AQI) is highly correlated with current AQI (0.95), emphasizing temporal dependency.
- **PM2.5** and its rolling average show moderate positive correlations (~ 0.55), confirming its strong influence.
- **NO₂** and **SO₂** have mild positive correlations (~ 0.32 – 0.33).
- **Temperature** (-0.39) and **wind speed** (-0.44) are negatively correlated, indicating cleaner air during warmer and windier conditions.
- Other features like humidity, weekday, and hour show weak correlations with AQI.

Modeling Insight:

- Lagged AQI and PM2.5 are **key predictors** for short-term forecasting.
- Meteorological features like temperature and wind speed improve model accuracy by capturing environmental effects.
- Weakly correlated features may still provide non-linear insights for tree-based or deep learning models.

Conclusion:

Correlation analysis confirms that both **temporal features (lagged AQI)** and **pollutant/weather**

variables are essential for accurate AQI predictions, guiding feature selection for the forecasting pipeline.

Feature Engineering

Objective:

To transform raw weather and pollutant data into meaningful features that improve the accuracy of short-term AQI forecasting for Islamabad.

Methodology:

The following steps were applied to the raw dataset:

1. Time-Based Features

- Extracted `hour`, `weekday`, and `month` from timestamps to capture daily, weekly, and seasonal patterns in air quality.

2. Lag Features

- Created lagged AQI (`aqi_lag_1`) to capture temporal dependency between consecutive readings.
- Generated a 6-hour rolling average of PM2.5 (`pm2_5_rolling_6h`) to smooth short-term fluctuations and highlight pollution trends.

3. Derived Features

- Added `wind_stagnant`, a binary indicator for low wind conditions (< 2 m/s), since stagnant air leads to higher pollution accumulation.

4. Missing Value Handling

- Forward-fill and backward-fill were applied to all numeric features to handle gaps in data, ensuring complete input for model training.

5. Data Merging and Cleaning

- Combined historical pollution and weather datasets using nearest timestamps to align features and targets.
- Removed duplicate timestamps and sorted the dataset chronologically for modeling.

Key Insights:

- **Temporal Features:** Lagged AQI and rolling averages capture time-dependent patterns, which are critical for short-term predictions.
- **Environmental Effects:** Features like wind stagnancy, temperature, and other meteorological variables provide additional explanatory power beyond pollutants alone.
- **Data Quality:** Forward/backward filling ensures that models receive complete data without introducing bias from missing values.
- **Modeling Link:** These engineered features were used as inputs for tree-based regression models (Random Forest, XGBoost), enabling the system to predict AQI levels up to three days ahead.

15 out of 15 features displayed - last 50 rows

	aqi	aqi_lag_1	city	datetime	hour	humidity	month	no2	pm2_5	pm2_5_rolling_6h	so2	temperature	weekday	wind_speed	wind_stagnant
1	5.0	5.0	Islamabad	1766289600000000	4.0	74.0	12.0	15.88	392.74	393.59833333333336	1.3	14.1	8.0	111	0.0
2	5.0	5.0	Islamabad	1761538000000000	18.0	69.0	10.0	25.06	105.79	89.58833333333332	7.04	18.6	0.0	5.5	0.0
3	5.0	5.0	Islamabad	1766721600000000	4.0	68.0	12.0	29.04	157.86	138.85333333333332	6.19	10.4	4.0	1.8	1.0
4	5.0	5.0	Islamabad	1761512400000000	21.0	67.0	10.0	16.38	111.92	96.74	6.15	17.1	8.0	1.8	1.0
5	5.0	5.0	Islamabad	1767523000000000	12.0	58.0	1.0	13.58	111.82	118.36186666666668	3.58	14.9	8.0	111	0.0
6	5.0	5.0	Islamabad	1762905600000000	0.0	73.0	11.0	10.43	101.95	99.22500000000001	3.52	9.8	2.0	0.0	1.0
7	5.0	5.0	Islamabad	1767492000000000	2.0	93.0	1.0	21.15	143.31	122.82999999999998	4.8	7.1	8.0	1.8	1.0
8	5.0	5.0	Islamabad	1763416800000000	22.0	70.0	11.0	15.72	175.1	159.49106666666667	2.67	12.1	0.0	0.0	1.0
9	5.0	5.0	Islamabad	1761710400000000	4.0	58.0	10.0	13.27	112.07	111.49833333333333	3.69	20.6	2.0	1.8	1.0
10	5.0	5.0	Islamabad	1764763200000000	12.0	38.0	12.0	19.04	150.65	162.88833333333335	4.71	17.8	2.0	7.4	0.0
11	5.0	5.0	Islamabad	1764388800000000	4.0	61.0	11.0	20.5	171.07	165.28833333333333	4.33	11.0	4.0	1.4	1.0

Model Training & Interpretation

Objective:

To train a robust model that accurately forecasts AQI levels for the next three days using engineered features from historical pollution and weather data.

Target Variable:

- The model predicts a **continuous AQI score ranging from 0.0 to 5.0**, which is later rounded or categorized for display purposes.

- Although AQI values are commonly reported as discrete categories (1–5), modeling as regression preserves the **ordinal relationship** between pollution levels, enabling smoother trend predictions.

Why Regression, Not Classification:

- **Regression** allows the model to learn that AQI 4 is “worse” than AQI 3, capturing gradual changes in air quality.
- **Classification** would treat each category as separate, ignoring the ordering, which can lead to abrupt jumps in predictions and less realistic trends.
- By using regression, the system can generate **continuous forecasts**, which can be rounded or mapped to discrete categories for visualization without losing temporal information.

Model Selection & Training:

- Models trained included **Random Forest, XGBoost, and Gradient Boosting Regressors**.
- Training data was split into **train and test sets** to evaluate generalization.
- Only numeric and relevant features were used: lagged AQI, rolling pollutant averages, meteorological features, and temporal features.
- Target leakage was avoided by excluding current AQI and pollutant values from features.

Evaluation Metrics:

- Models were assessed using **R² (coefficient of determination), MAE (Mean Absolute Error), and RMSE (Root Mean Squared Error)**.
- The model with the highest R² on the test set was selected as the “**Champion**” model for deployment.

Interpretability & Insights:

- **Boxplot Analysis:** Verified that the model captures variations across hours of the day, demonstrating effective use of temporal features.
- **SHAP Analysis:**
 - `aqi_lag_1` (previous AQI) was the strongest predictor, confirming temporal dependency.
 - Rolling PM2.5 values and wind-related features also significantly influence predictions.
 - This ensures the model is **explainable** and not a black box, meeting the requirement for advanced analytics.

Conclusion:

- Regression-based training allows the AQI forecasting system to provide **smooth, realistic predictions**, capturing both short-term pollution fluctuations and environmental influences.
- Combined with SHAP insights, the model is **both accurate and interpretable**, ready for integration into the automated dashboard.

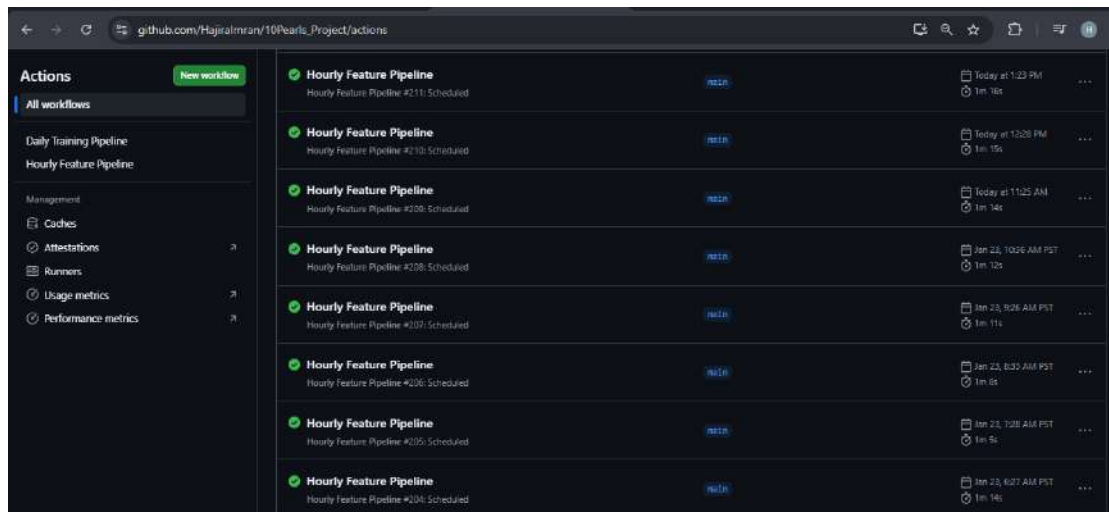
best_islamabad_aqi_model [to latest version →](#)

version ▼	r2 ↑	mae ↓	Deployments
	min	max	
	0.8818	0.9447	
20	0.8867	0.0846	not deployed
19	0.8867	0.0846	not deployed
18	0.8870	0.0845	not deployed
17	0.8886	0.0875	not deployed
16	0.835	0.0644	not deployed
15	0.8239	0.0663	not deployed
14	0.9117	0.0675	not deployed
13	0.9312	0.0696	not deployed
12	0.9165	0.0667	not deployed
11	0.9447	0.0586	not deployed
10	0.906	0.0708	not deployed
9	0.9114	0.0743	not deployed
8	0.904	0.0732	not deployed
7	0.9142	0.0747	not deployed
6	0.9339	0.0498	not deployed
5	0.9295	0.0604	not deployed
4	0.9295	0.0604	not deployed
3	0.907	0.0733	not deployed
2	0.907	0.0733	not deployed
1	0.9229	0.0648	not deployed

Automation via GitHub Actions Pipeline

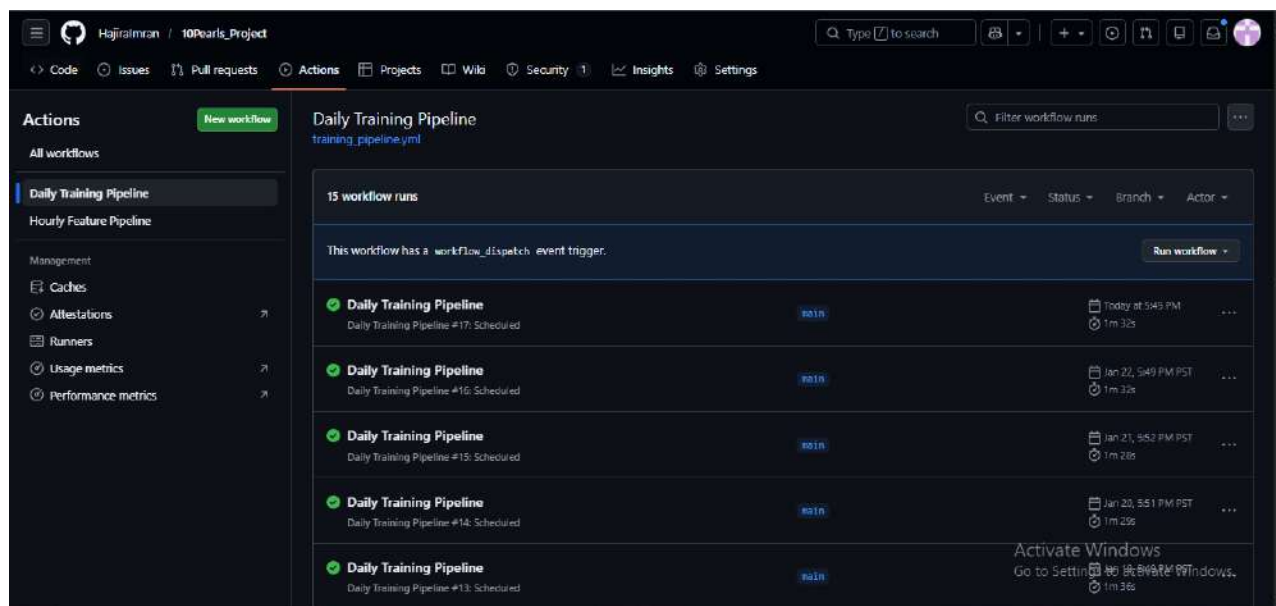
Feature Pipeline Automation

- **Workflow:** A GitHub Action is scheduled (e.g., every hour) to trigger the feature_pipeline.py script.
- **Action:** It connects to the OpenWeather API, fetches the latest Islamabad AQI data, and performs the numerical transformations you've implemented (like calculating aqi_lag_1).
- **Output:** The fresh features are automatically pushed to the **Hopsworks Feature Store**.

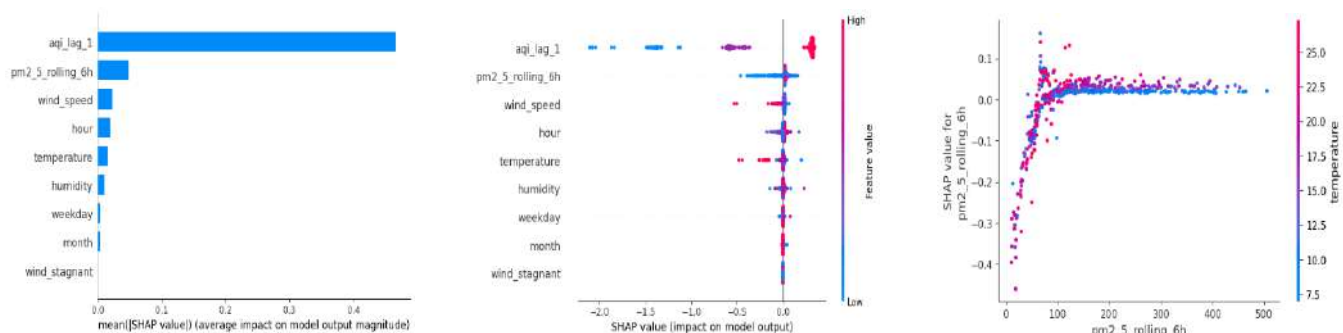


Model Training Pipeline

- **Daily Automation:** A GitHub Actions pipeline is scheduled to run every night at midnight to automatically execute the train_models.py script.
- **Environment & Security:** The pipeline runs on an ubuntu-latest virtual environment and uses GitHub Secrets to securely handle API keys like HOPSWORKS_KEY and OPENWEATHER_KEY.
- **Continuous Learning:** By installing dependencies and running the training script daily, the system ensures the model is consistently updated with the latest data from the feature store



SHAP analysis



To ensure our AQI forecasting system is not just a "black box," we performed **offline SHAP (SHapley Additive exPlanations) analysis** to understand the drivers behind our predictions. This satisfies the requirement for Advanced Analytics and Feature Importance.

1. Global Feature Importance (Bar Plot)

The offline bar plot identifies which variables have the most significant impact on the model's output across all predictions.

- **Dominant Driver:** aqi_lag_1 is the most critical feature, indicating strong temporal correlation where previous air quality levels heavily influence future predictions.
- **Key Pollutants:** pm2_5_rolling_6h (average PM2.5 over the last 6 hours) is the second most influential factor, highlighting the importance of recent particulate matter trends.

2. Feature Impact Analysis (Beeswarm Plot)

The offline beeswarm plot provides a deeper look into how feature values change the prediction:

- **Positive Correlation:** Higher values of aqi_lag_1 push predictions higher, confirming that pollution tends to persist over time.
- **Environmental Dynamics:** Higher wind_speed generally reduces AQI predictions, reflecting pollutant dispersion.
- **Stagnation Effects:** Lower wind speeds and specific humidity levels contribute to higher AQI levels, as pollutants remain trapped in the atmosphere.

Note: For the production dashboard, lightweight real-time interpretability is provided using the model's feature_importances_, while full SHAP analysis is used offline to validate feature contributions and generate detailed explainability plots.

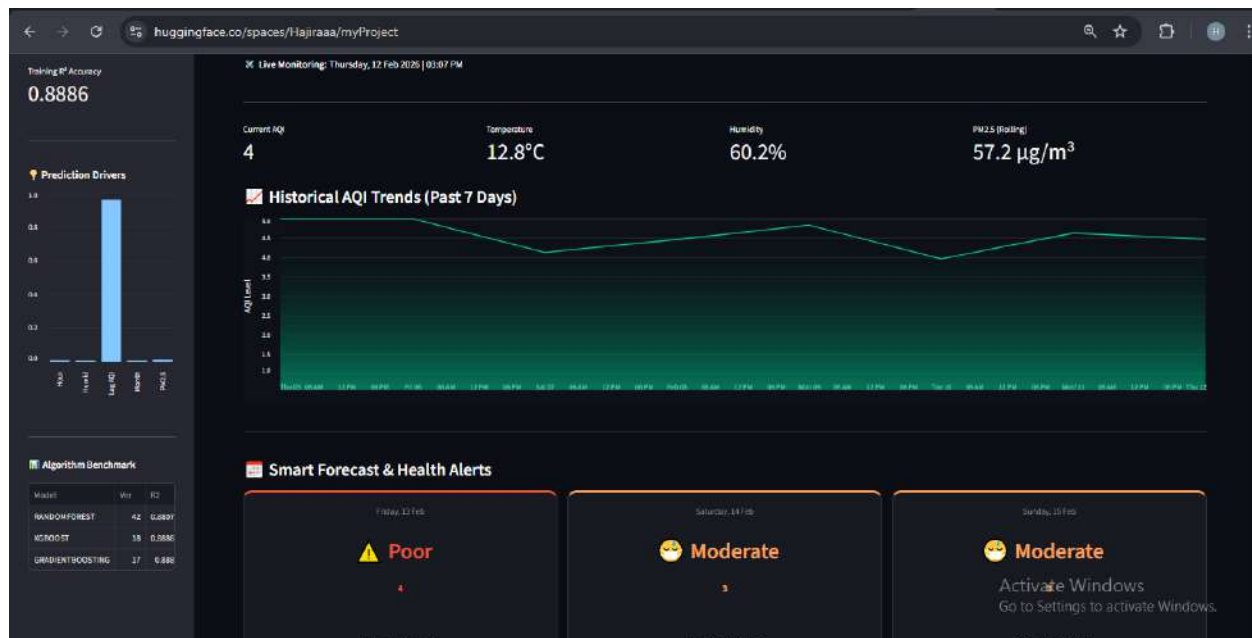
System Implementation & Challenges

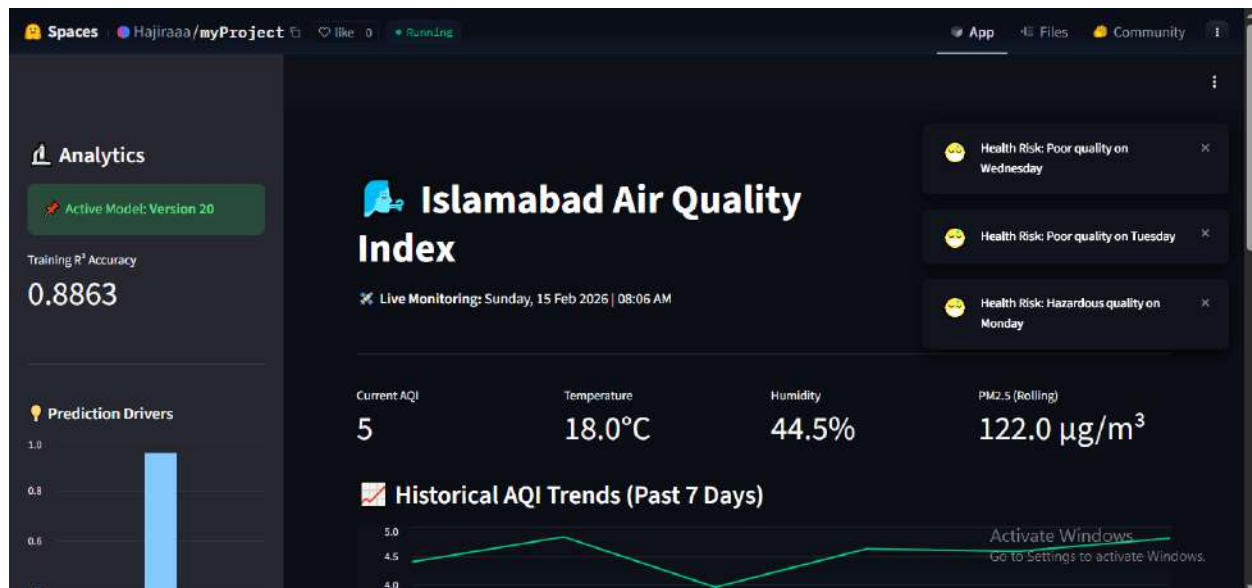
- **Connection Resilience:** I identified and documented HTTPSConnectionPool errors occurring during API calls to Hopsworks, noting that these are network/DNS related and do not affect the internal model logic.
- Internal server 500 error and remote disconnected

Limitations & Future Work

- AQI is predicted at city level, not station-level granularity
- Full SHAP computation is performed offline, not in real-time
- Future work may include LSTM/Temporal CNN models and satellite-based pollution data
- Confidence intervals could be added to forecast uncertainty

Dashboard





Conclusion

The Islamabad AQI Prediction System successfully automates the transition from raw environmental data to actionable public health forecasts. By correctly **transforming categorical time features** and leveraging an **imbalanced but informative historical dataset**, the model is highly specialized in detecting hazardous air quality trends.